

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0124

Pentland 138 kV
May 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0124. The Interconnection Customer proposes a 141.4 MW solar photovoltaic facility interconnecting at the Pentland 138 kV substation in the Virginia area. The request was received on April 11, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 81 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,271,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Person / State School 345.0 kV L is overloaded to 114.1% of its 1,605 MVA emergency rating for the loss of the Person Substation / Mecklenburg Power Station 345.0 kV Line. The project contributes 12.9% DFAX (18.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
State School / Person Substation 345.0 kV	Rebuild line with higher-capacity conductor	\$51,046,000	6% (\$3,271,000)
	Total allocated to project		\$3,271,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.